PERMANENCE OF LOCAL FINITE PRESENTATION/TYPE ALONG FPPF COVERS WITH SURJECTIVE FLAT LFP F

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites three Morphisms of Spaces lemmas and one Descent lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Local finite type/presentation is \'etale local on the source. Stacks Project, Morphisms of Spaces, Lemmas spaces-morphisms-lemma-finite-type-local and spaces-morphisms-lemma-finite-presentation-local. Justifies the reduction to affine B and Y.
- Openness of fppf morphisms on points. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-fppf-open. Lets the proof pick U with $U \to Y$ surjective.
- Scheme-side permanence. Stacks Project, Descent, Lemma descent-lemma-flat-finitely-presented-permanence. Concludes for the scheme diagram.

Source reference. Original reference 0469: sources/stacks-chow/tags.txt, source lines 5464–5464. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 06NR: sources/stacks-chow/tags.txt, source lines 8545–8545. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 040Y: sources/stacks-chow/tags.txt, source lines 5273–5273. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

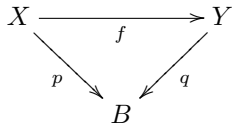
Source reference. Original reference 0410: sources/stacks-chow/tags.txt, source lines 5275–5275. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 042S: sources/stacks-chow/tags.txt, source lines 5339–5339. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02KL: sources/stacks-chow/tags.txt, source lines 3376–3376. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:spaces-descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let*



be a commutative diagram of morphisms of algebraic spaces over S. Assume that f is surjective, flat, and locally of finite presentation and assume that p is locally of finite presentation (resp. locally of finite type). Then q is locally of finite presentation (resp. locally of finite type).

Proof. Since $\{X \to Y\}$ is an fppf covering, it induces a surjection of fppf sheaves (Topologies on Spaces, Lemma 0469) and the lemma is a special case of Lemma 06NR. On the other hand, an easier argument is to deduce it from the analogue for schemes. Namely, the problem is étale local on B and Y (Morphisms of Spaces, Lemmas 040Y and 0410). Hence we may assume that B and Y are affine schemes. Since $|X| \to |Y|$ is open (Morphisms of Spaces, Lemma 042S), we can choose an affine scheme U and an étale morphism $U \to X$ such that the composition $U \to Y$ is surjective. In this case the result follows from Descent, Lemma 02KL. □